

HYPERPARAMETER SELECTION FOR HIGH DIMENSIONAL SPARSE LEARNING: APPLICATION TO NEUROIMAGING

A GENTLE MOTIVATION

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**UNIVERSITÉ DE
MONTPELLIER**



JOINT WORKS WITH VARIOUS COLLEAGUES



Quentin Bertrand (MILA)

Quentin Klopfenstein (Université du Luxembourg)

Mathurin Massias (INRIA, OCKHAM them)

Pierre-Antoine Bannier (M2 student, Parietal Team)

Samuel Vaiter (Université Côte d'Azur, CNRS)

Mathieu Blondel (Google Research, Brain team)

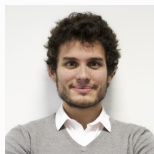
Alexandre Gramfort (INRIA, Parietal Team)



Quentin B.



Quentin K.



Mathurin



Pierre-Antoine



Mathieu



Samuel



Alexandre

NEUROIMAGING DATA: EEG ⁽¹⁾ AND MEG ⁽²⁾



(a) EEG



(b) MEG=Mag.+Grad.



(c) M/EEG

Photo credit: S. Whitmarsh

⁽¹⁾ H. Berger (1929). "Über das elektroenkephalogramm des menschen". In: *Archiv für psychiatrie und nervenkrankheiten*

⁽²⁾ D. Cohen (1968). "Magnetoencephalography: evidence of magnetic fields produced by alpha-rhythm currents". In: *Science*

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- **Data Y:** electric and magnetic fields at the head surface

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NEUROIMAGING DATA: EEG ⁽¹⁾ AND MEG ⁽²⁾



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- ▶ **Data** Y : electric and magnetic fields at the head surface
- ▶ **Goal**: which parts of the brain are responsible for the signals?

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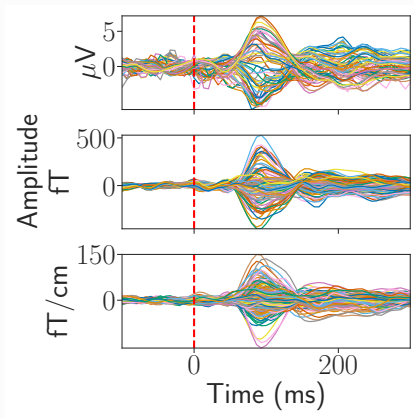
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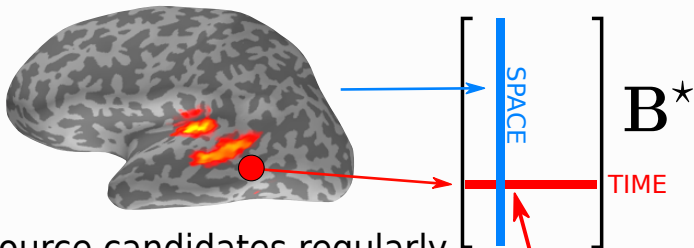
- ▶ **Data** Y : electric and magnetic fields at the head surface
- ▶ **Goal**: which parts of the brain are responsible for the signals?
- ▶ **Applications**: clinical and cognitive experiments

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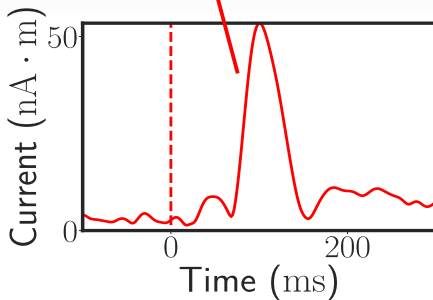


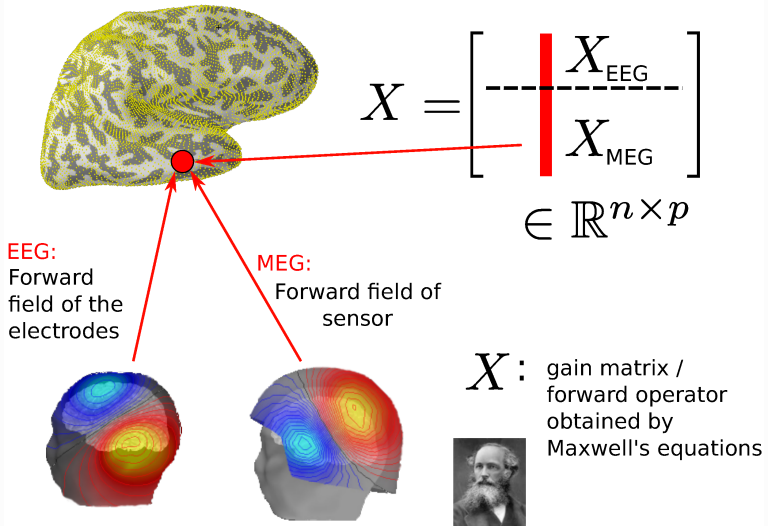
- 3 modalities:
- ▶ EEG
 - ▶ MEG: magnometers (amplitude)
 - ▶ MEG: gradiometers (gradients)



Source candidates regularly spaced in the brain (e.g., every 5mm)

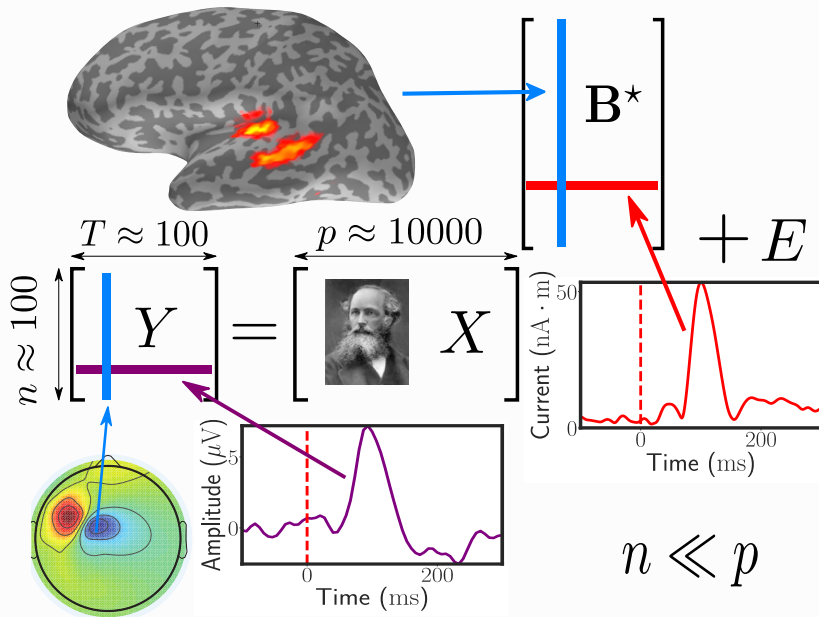
$$B^* \in \mathbb{R}^{p \times T}$$





THE M/EEG INVERSE PROBLEM

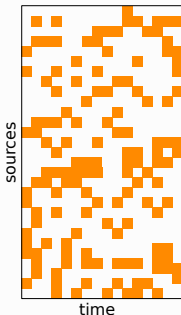
MAXWELL EQUATIONS AND (APPROX.) LINEARITY





Popular convex penalties:

$$\hat{\mathbf{B}} \in \arg \min_{\mathbf{B} \in \mathbb{R}^{p \times T}} \left(\frac{1}{2nT} \|\mathbf{Y} - \mathbf{X}\mathbf{B}\|^2 + \lambda \Omega(\mathbf{B}) \right)$$



Sparse support: no structure

Penalty: **Lasso**

$$\Omega(\mathbf{B}) = \|\mathbf{B}\|_1 = \sum_{j=1}^p \sum_{k=1}^T |\mathbf{B}_{j,k}|$$

Parameter $\hat{\mathbf{B}} \in \mathbb{R}^{p \times T}$

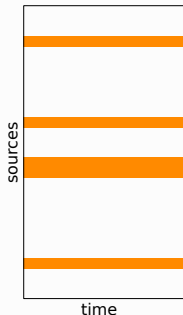
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Popular convex penalties: multitask Lasso (MTL)

$$\hat{\mathbf{B}} \in \arg \min_{\mathbf{B} \in \mathbb{R}^{p \times T}} \left(\frac{1}{2nT} \|\mathbf{Y} - \mathbf{X}\mathbf{B}\|^2 + \lambda \Omega(\mathbf{B}) \right)$$



Parameter $\hat{\mathbf{B}} \in \mathbb{R}^{p \times T}$

Sparse support: group structure ✓

Penalty: **Group-Lasso**

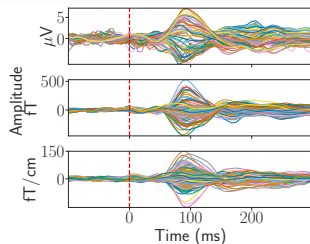
$$\Omega(\mathbf{B}) = \|\mathbf{B}\|_{2,1} = \sum_{j=1}^p \|\mathbf{B}_{j,:}\|_2$$

where $\mathbf{B}_{j,:}$: the j -th row of $\mathbf{B}$

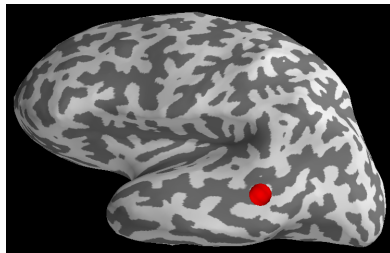
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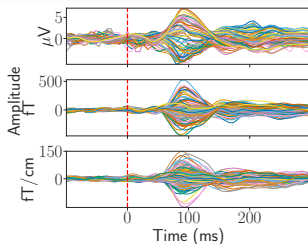
SUMMARY OF THE PROBLEM SETTING



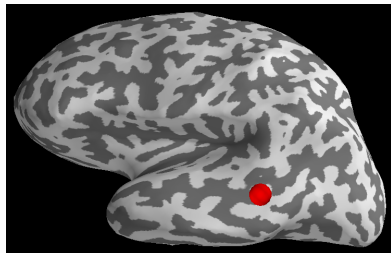
What you have: $Y \in \mathbb{R}^{n \times T}$



What you want: $B \in \mathbb{R}^{p \times T}$



What you have: $\mathbf{Y} \in \mathbb{R}^{n \times T}$



What you want: $\mathbf{B} \in \mathbb{R}^{p \times T}$

This is typically done using optimization based estimators

$$\hat{\mathbf{B}} \in \arg \min_{\mathbf{B} \in \mathbb{R}^{p \times T}} \left(\frac{1}{2nT} \|\mathbf{Y} - \mathbf{XB}\|_F^2 + \lambda \Omega(\mathbf{B}) \right)$$



$$\hat{\mathbf{B}} \in \arg \min_{\mathbf{B} \in \mathbb{R}^{p \times T}} \left(\frac{1}{2nT} \|\mathbf{Y} - \mathbf{XB}\|_F^2 + \lambda \Omega(\mathbf{B}) \right)$$

Covered in this presentation

- ▶ How to efficiently select the regularization parameter λ ?^{(1), (2)}

Not covered in this presentation

- ▶ How to efficiently solve this optimization problem?⁽³⁾
- ▶ How to handle spatial correlation?⁽⁴⁾

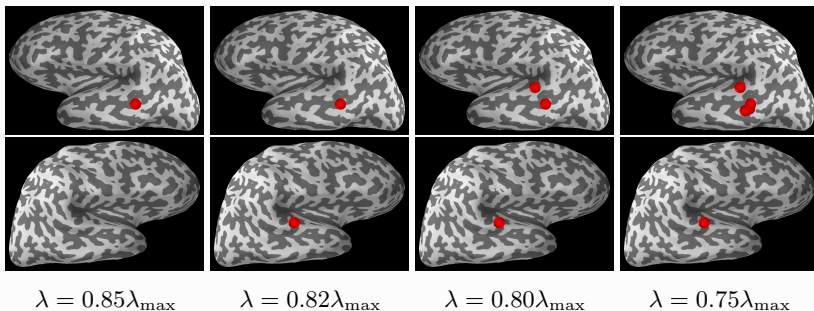
⁽¹⁾ Q Bertrand, Q Klopfenstein, M. Blondel, et al. (2020). "Implicit differentiation of Lasso-type models for hyperparameter optimization". In: *ICML*.

⁽²⁾ Q Bertrand, Q Klopfenstein, M. Massias, et al. (2021). "Implicit differentiation for fast hyperparameter selection in non-smooth convex learning". In: *Submitted to JMLR*.

⁽³⁾ Q Bertrand and M. Massias (2021). "Anderson acceleration of coordinate descent". In: *AISTATS*.

⁽⁴⁾ Q Bertrand, M. Massias, et al. (2019). "Handling correlated and repeated measurements with the smoothed Multivariate square-root Lasso". In: *NeurIPS*.

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Real M/EEG data. Brain source reconstruction using multitask Lasso with multiple λ . Which λ to pick? How to *automatically* select λ ?

- When $\lambda \geq \lambda_{\max}$, $\hat{\mathbf{B}} = 0$ no sources are recovered



- ▶ Statistical route^{(1), (2)}: assumptions on the design matrix X , provide guarantees but are often too
- ▶ Bayesian statistics^{(3), (4)}: prior on λ
- ▶ Bayesian optimisation,⁽⁵⁾ 0-th order method:

The road today:

- ▶ Hyperparameter optimization⁽⁶⁾: minimize a given criterion $\mathcal{C}(\hat{\beta}^{(\lambda)})$

⁽¹⁾ K. Lounici (2008). "Sup-norm convergence rate and sign concentration property of Lasso and Dantzig estimators". In: *Electron. J. Stat.*

⁽²⁾ K. Lounici, M. Pontil, et al. (2009). "Taking Advantage of Sparsity in Multi-Task Learning". In: *arXiv preprint arXiv:0903.1468*.

⁽³⁾ M. E. Tipping (2001). "Sparse Bayesian learning and the relevance vector machine". In: *Journal of Machine Learning Research*.

⁽⁴⁾ M. Figueiredo (2001). "Adaptive Sparseness Using Jeffreys Prior.". In: *Advances in Neural Information Processing Systems*.

⁽⁵⁾ F. Hutter, J. Lücke, and L. Schmidt-Thieme (2015). "Beyond manual tuning of hyperparameters". In: *KI-Künstliche Intelligenz*.

⁽⁶⁾ R. Kohavi and G. H. John (1995). "Automatic parameter selection by minimizing estimated error". In: *Machine Learning Proceedings*.

Possible selection criterion:

- ▶ Good generalization^{(1), (2)} of $\hat{\beta}(\lambda)$
- ▶ AIC/BIC,⁽³⁾ SURE⁽⁴⁾ that controls model complexity

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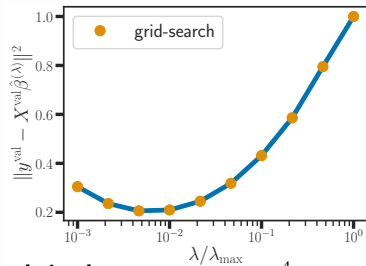
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- ▶ Good generalization^{(1), (2)} of $\hat{\beta}(\lambda)$
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Real-sim dataset, $n \approx p \approx 10^4$
 Validation loss as a function of λ .

Simplified example ($T = 1$):

Model: Lasso

$$\hat{\beta}(\lambda) \in \arg \min_{\beta \in \mathbb{R}^p} \frac{\|y^{\text{train}} - X^{\text{train}} \beta\|^2}{2n} + \lambda \|\beta\|_1$$

Criterion: held-out loss

$$\arg \min_{\lambda} \|y^{\text{val}} - X^{\text{val}} \hat{\beta}(\lambda)\|^2$$

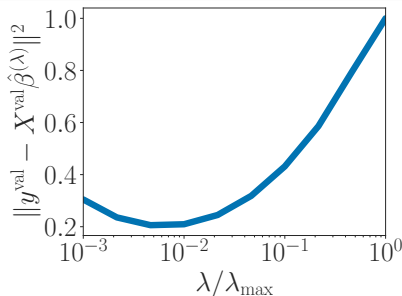
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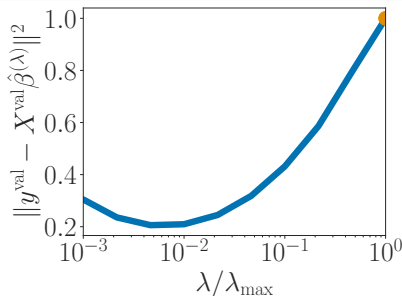
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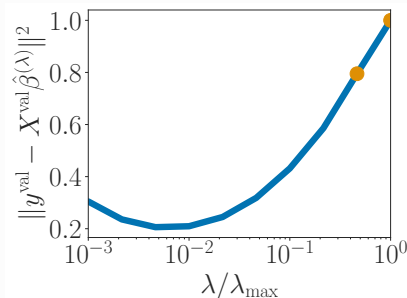
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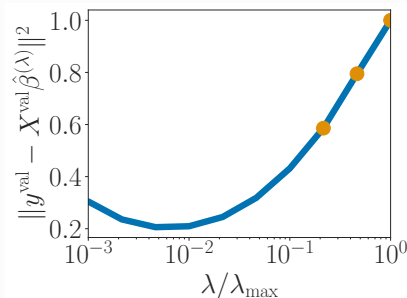
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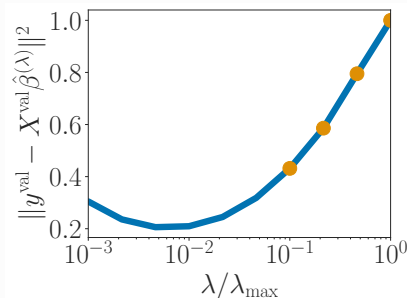
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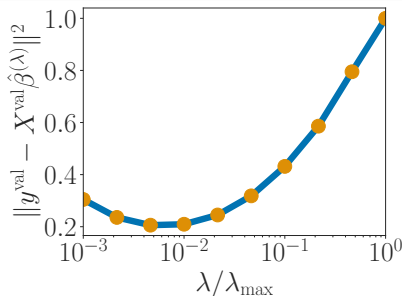


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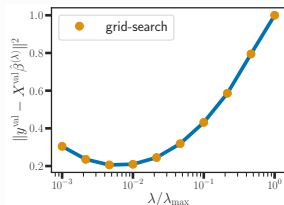


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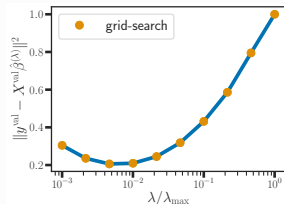
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- Grid-search, random-search,⁽¹⁾ SMBO⁽²⁾:
0-order methods to solve bilevel optimization problem

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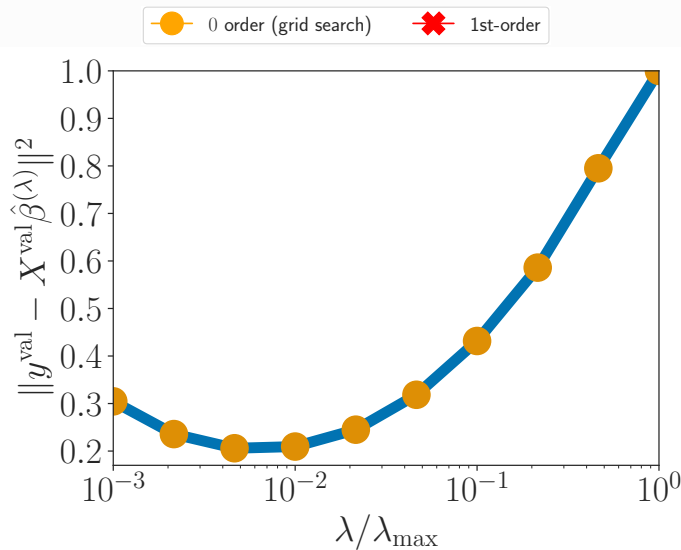
- ▶ Grid-search, random-search,⁽¹⁾ SMBO⁽²⁾:
0-order methods to solve bilevel optimization problem
- ▶ **Idea:** if $\mathcal{L}$ is differentiable, use 1st-order optimization
 - ▶ Compute gradient: $\nabla_{\lambda} \mathcal{L}$
 - ▶ Perform gradient descent step⁽³⁾:

$$\lambda^{(t+1)} = \lambda^{(t)} - \rho \nabla_{\lambda} \mathcal{L}(\lambda^{(t)}) \quad \text{with } \rho > 0$$

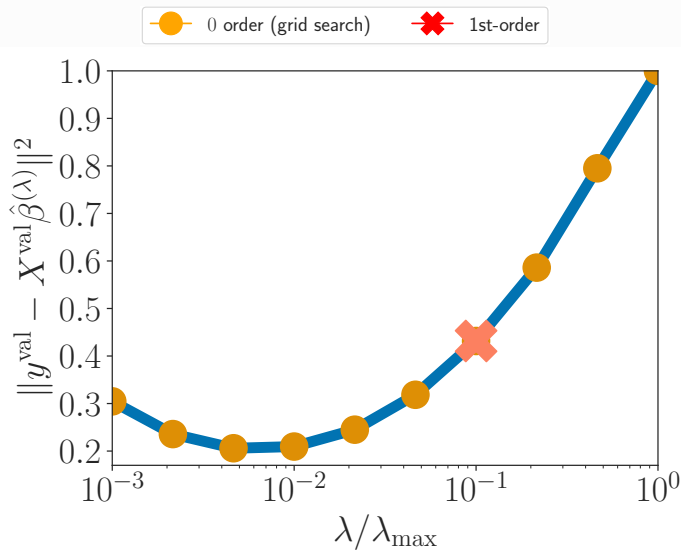
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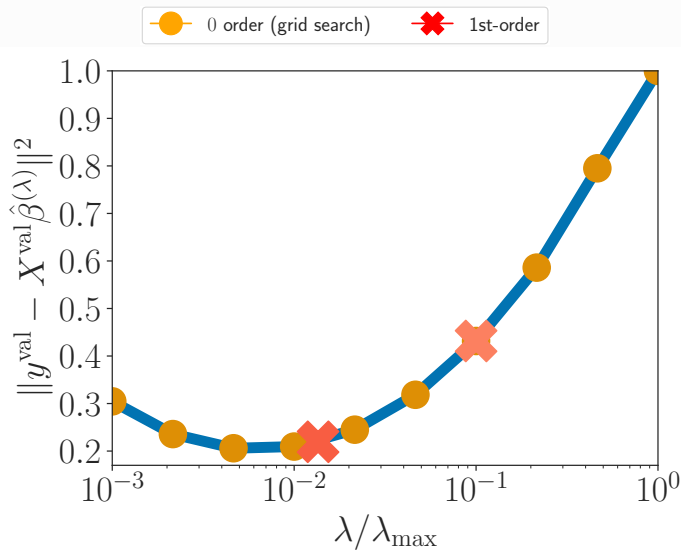
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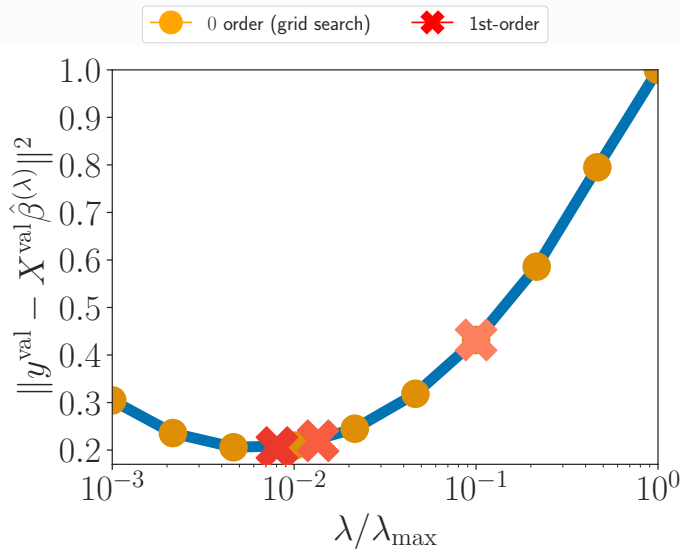
Real-sim dataset, $n \approx p \approx 10^4$. Validation loss as a function of λ .



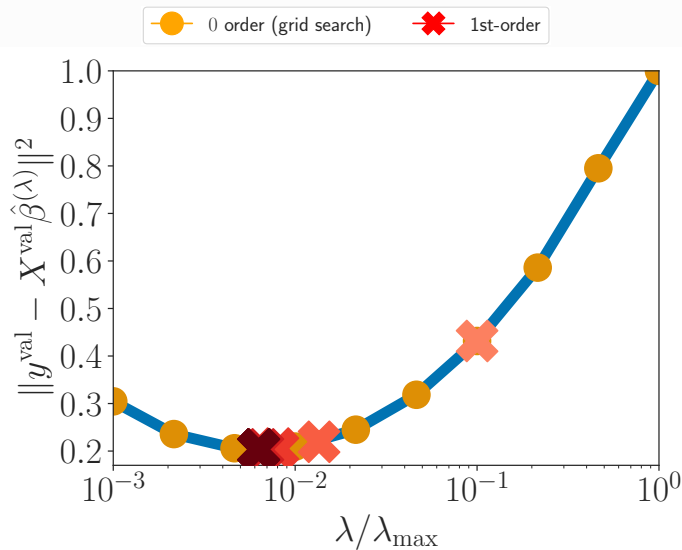
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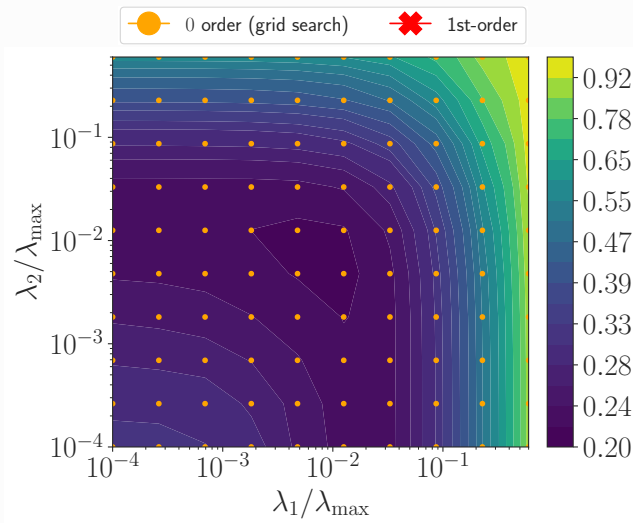
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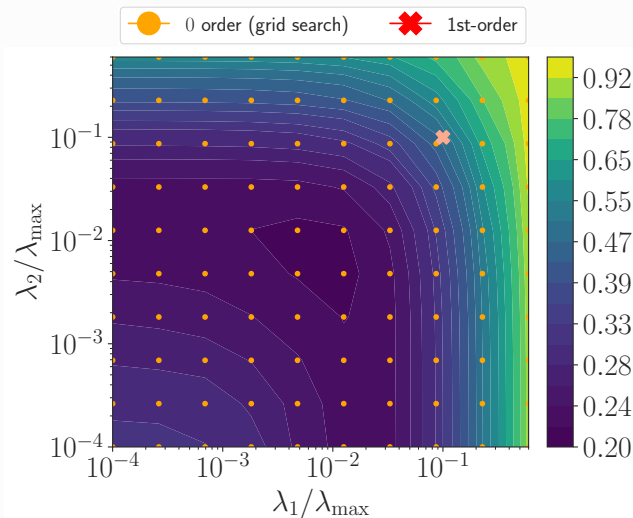


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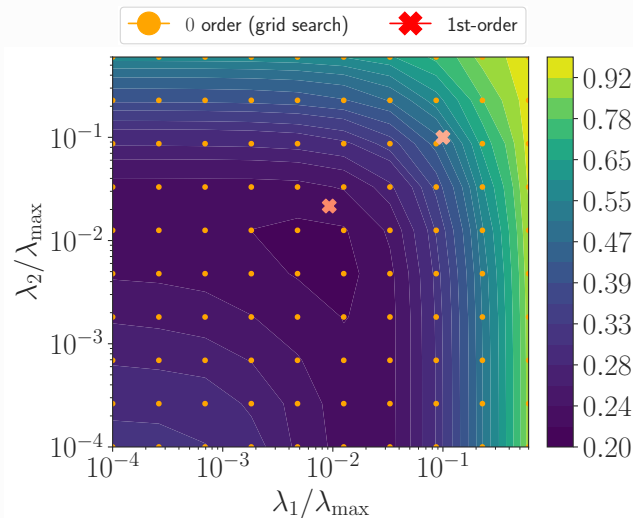
Real-sim dataset, level sets of the validation loss (hold-out)

$$\arg \min_{\beta} \frac{1}{2n} \|\mathbf{y}^{\text{train}} - \mathbf{X}^{\text{train}} \beta\|^2 + \lambda_1 \|\beta\|_1 + \frac{\lambda_2}{2} \|\beta\|^2$$



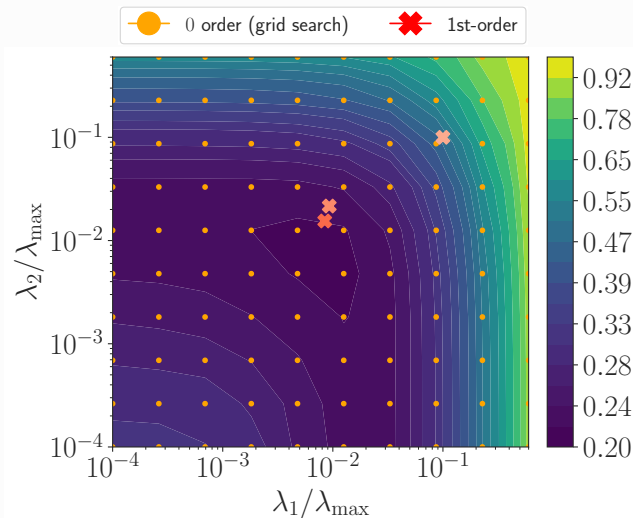
Real-sim dataset, level sets of the validation loss (hold-out)

$$\arg \min_{\beta} \frac{1}{2n} \| \mathbf{y}^{\text{train}} - \mathbf{X}^{\text{train}} \beta \|^2 + \lambda_1 \|\beta\|_1 + \frac{\lambda_2}{2} \|\beta\|^2$$



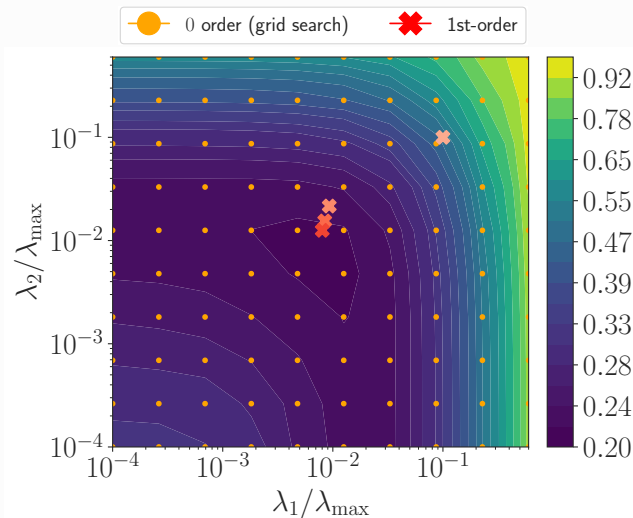
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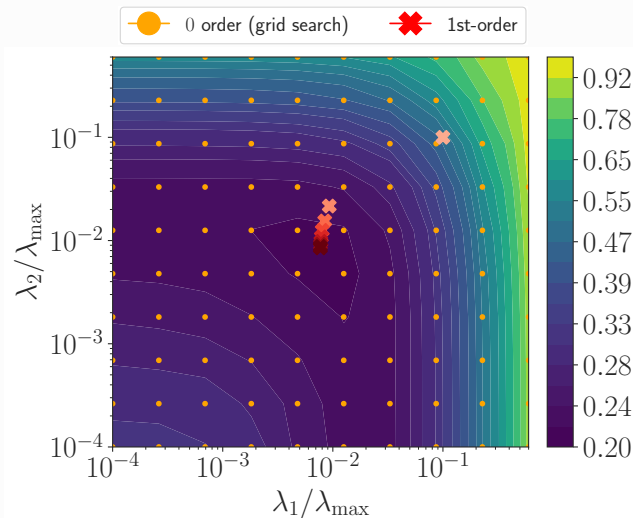
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Real-sim dataset, level sets of the validation loss (hold-out)

$$\arg \min_{\beta} \frac{1}{2n} \|\mathbf{y}^{\text{train}} - \mathbf{X}^{\text{train}} \beta\|^2 + \lambda_1 \|\beta\|_1 + \frac{\lambda_2}{2} \|\beta\|^2$$

$$\begin{aligned} \arg \min_{\lambda \in \mathbb{R}} \left\{ \mathcal{L}(\lambda) := C(\hat{\beta}^{(\lambda)}) := \|\mathbf{y}^{\text{val}} - \mathbf{X}^{\text{val}} \hat{\beta}^{(\lambda)}\|^2 \right\} \\ \text{s.t. } \hat{\beta}^{(\lambda)} \in \arg \min_{\beta \in \mathbb{R}^p} \frac{1}{2n} \|\mathbf{y}^{\text{train}} - \mathbf{X}^{\text{train}} \beta\|^2 + \lambda \|\beta\|_1 \end{aligned}$$

⁽¹⁾ J. Nocedal and S. J. Wright (2006). *Numerical optimization*. Second. Springer Series in Operations Research and Financial Engineering. Springer, Chap. 3.

⁽²⁾ D. C. Liu and J. Nocedal (1989). "On the limited memory BFGS method for large scale optimization". In: *Mathematical programming*.

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Once $\nabla_{\lambda} \mathcal{L}(\lambda)$ is computed, one can use standard first-order methods:

- ▶ Line-search⁽¹⁾
- ▶ L-BFGS⁽²⁾
- ▶ Gradient descent

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Once $\nabla_{\lambda} \mathcal{L}(\lambda)$ is computed, one can use standard first-order methods:

- ▶ Line-search⁽¹⁾
- ▶ L-BFGS⁽²⁾
- ▶ Gradient descent

Main contribution here: compute $\nabla_{\lambda} \mathcal{L}(\lambda)$ for a given λ

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$$\begin{aligned} \arg \min_{\lambda \in \mathbb{R}} \left\{ \mathcal{L}(\lambda) := C(\hat{\beta}^{(\lambda)}) := \|y^{\text{val}} - X^{\text{val}} \hat{\beta}^{(\lambda)}\|^2 \right\} \\ \text{s.t. } \hat{\beta}^{(\lambda)} \in \arg \min_{\beta \in \mathbb{R}^p} \frac{1}{2n} \|y^{\text{train}} - X^{\text{train}} \beta\|^2 + \lambda \|\beta\|_1 \end{aligned}$$

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Chain rule:

$$\begin{aligned} \nabla_{\lambda} \mathcal{L}(\lambda) = \underbrace{\hat{\mathcal{J}}_{(\lambda)}^{\top}}_{\substack{:= (\nabla_{\lambda} \hat{\beta}_1^{(\lambda)}, \dots, \nabla_{\lambda} \hat{\beta}_p^{(\lambda)}) \\ \rightarrow \text{main challenge}}} \nabla_{\beta} C(\hat{\beta}^{(\lambda)}) \end{aligned}$$

► Boils down to:

how to compute the Jacobian $\hat{\mathcal{J}}_{(\lambda)} \in \mathbb{R}^{p \times 1}$ efficiently?

FORWARD-MODE DIFFERENTIATION ^{(1),(2)} OF PGD

PROXIMAL GRADIENT DESCENT ⁽³⁾



$$\hat{\beta}^{(\lambda)} \in \arg \min_{\beta \in \mathbb{R}^p} \underbrace{f}_{\text{smooth}}(\beta) + \lambda \underbrace{g}_{\text{non-smooth}}(\beta)$$

Algorithm: Proximal gradient descent PGD

```
init   :  $\beta = 0_p, \quad , L$ 
for iter = 1, ..., do
     $z \leftarrow \beta - \frac{1}{L} \nabla f(\beta)$                                      // gradient step
     $\beta \leftarrow \text{prox}_{\lambda g/L}(z)$                                 // proximal step: thresholding for us
return  $\beta$ 
```

⁽¹⁾ R. E. Wengert (1964). "A simple automatic derivative evaluation program". In: *Communications of the ACM* 7.8, pp. 463–464

⁽²⁾ C.-A. Deledalle et al. (2014). "Stein Unbiased GrAdient estimator of the Risk (SUGAR) for multiple parameter selection". In: *SIAM J. Imaging Sci.*

⁽³⁾ B. Martinet (1970). "Brève communication. Régularisation d'inéquations variationnelles par approximations successives". In: *Revue française d'informatique et de recherche opérationnelle. Série rouge* 4.R3, pp. 154–158

FORWARD-MODE DIFFERENTIATION ^{(1),(2)} OF PGD

PROXIMAL GRADIENT DESCENT ⁽³⁾



$$\hat{\beta}^{(\lambda)} \in \arg \min_{\beta \in \mathbb{R}^p} \overbrace{f}^{\text{smooth}}(\beta) + \lambda \overbrace{g}^{\text{non-smooth}}(\beta)$$

Algorithm: Forward-mode differentiation of PGD

```
init   :  $\beta = 0_p, \mathcal{J} = 0_p, L$  (Lipschitz-constant for  $f$ )  
for iter = 1, ..., do  
     $z \leftarrow \beta - \frac{1}{L} \nabla f(\beta)$  // gradient step  
     $d\mathcal{J} \leftarrow (\text{Id}_p - \frac{1}{L} \nabla^2 f(\beta)) \mathcal{J}$  // diff w.r.t.  $\lambda$ : chain rule  
     $\beta \leftarrow \text{prox}_{\lambda g/L}(z)$  // proximal step: thresholding for us  
  
return  $\beta$ 
```

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     $\beta \leftarrow \text{prox}_{\lambda g/L}(z)$  // proximal step: thresholding for us
     $\mathcal{J} \leftarrow \partial_z \text{prox}_{\lambda g/L}(z) dz$  // diff w.r.t.  $\lambda$ : chain rule
return  $\beta, \mathcal{J}$ 
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```
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            $+ \partial_\lambda \text{prox}_{\lambda g/L}(z)$  // do not forget this term!  
return  $\beta, \mathcal{J}$ 
```

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Motivation for proximal operators:

smooth non-smooth functions (sic),
e.g., to perform gradient descent

$$\text{prox}_{\frac{\lambda}{L}g}(z) = \arg \min_{z' \in \mathbb{R}^p} \frac{\lambda}{L}g(z') + \frac{1}{2} \|z' - z\|^2$$

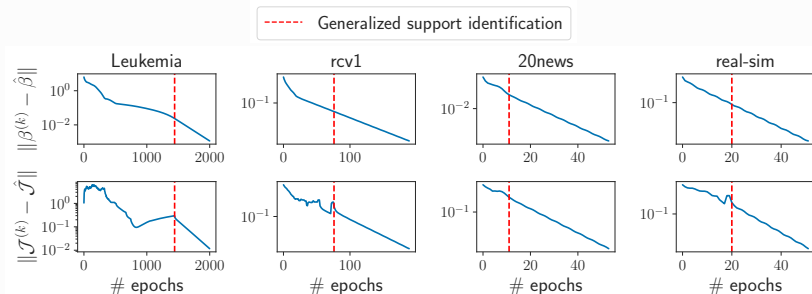


Jean-Jacques Moreau

⁽³⁾ J.-J. Moreau (1962). "Fonctions convexes duales et points proximaux dans un espace hilbertien". In: C. R. Acad. Sci. Paris 255, pp. 2897–2899

Forward diff. PCD convergence, Lasso

Provided that X (the design matrix) is not pathological, the sequence generated by PCD is converging to $\hat{\beta}$, and the Jacobian sequence based on forward differentiation converges to the true Jacobian. Moreover, once the support (the non-zeros coefs.) has been identified, convergence is linear.⁽¹⁾



⁽¹⁾ Q. Bertrand, Q. Klopfenstein, M. Blondel, et al. (2020). "Implicit differentiation of Lasso-type models for hyperparameter optimization". In: *ICML*



$$\hat{\beta}^{(\lambda)} \in \arg \min_{\beta \in \mathbb{R}^p} \psi(\beta, \lambda)$$

⁽¹⁾ Y. Bengio (2000). "Gradient-based optimization of hyperparameters". In: *Neural computation*



$$\hat{\beta}^{(\lambda)} \in \arg \min_{\beta \in \mathbb{R}^p} \psi(\beta, \lambda)$$

$$\nabla_{\beta} \psi(\hat{\beta}^{(\lambda)}, \lambda) = 0$$

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$$\nabla_{\beta, \lambda}^2 \psi(\hat{\beta}^{(\lambda)}, \lambda) + \hat{\mathcal{J}}_{(\lambda)}^{\top} \nabla_{\beta}^2 \psi(\hat{\beta}^{(\lambda)}, \lambda) = 0$$

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$$\nabla_{\beta, \lambda}^2 \psi(\hat{\beta}^{(\lambda)}, \lambda) + \hat{\mathcal{J}}_{(\lambda)}^{\top} \nabla_{\beta}^2 \psi(\hat{\beta}^{(\lambda)}, \lambda) = 0$$

$$\hat{\mathcal{J}}_{(\lambda)}^{\top} = -\nabla_{\beta, \lambda}^2 \psi(\hat{\beta}^{(\lambda)}, \lambda) \underbrace{\left(\nabla_{\beta}^2 \psi(\hat{\beta}^{(\lambda)}, \lambda) \right)^{-1}}_{p \times p}$$

- Need to solve a linear **system of size p**
- Prohibitive for large p

⁽¹⁾ Y. Bengio (2000). "Gradient-based optimization of hyperparameters". In: *Neural computation*



General formulation:

- ▶ Solve a linear **system of size p**
- ▶ Prohibitive for large p

With a sparsity inducing penalty:

- ▶ Solve a linear **system of size S** (sparsity degree of the estimator)
- ▶ $S \ll p$ very often



$$\hat{\beta}^{(\lambda)} = \text{ST} \left(\hat{\beta}^{(\lambda)} - \frac{1}{L} \nabla f(\hat{\beta}^{(\lambda)}), \frac{\lambda}{L} \right)$$

⁽¹⁾ Q. Bertrand, Q. Klopfenstein, M. Massias, et al. (2021). "Implicit differentiation for fast hyperparameter selection in non-smooth convex learning".
In: *Submitted to JMLR*



$$\hat{\beta}^{(\lambda)} = \text{ST} \left(\hat{\beta}^{(\lambda)} - \frac{1}{L} \nabla f(\hat{\beta}^{(\lambda)}), \frac{\lambda}{L} \right)$$

$$\begin{aligned} \hat{\mathcal{J}} &= \partial_{\beta} \text{ST} \left(\hat{\beta}^{(\lambda)} - \frac{1}{L} \nabla f(\hat{\beta}^{(\lambda)}), \frac{\lambda}{L} \right) \\ &+ \left(\text{Id} - \frac{\nabla^2 f}{L} \right) \hat{\mathcal{J}} + \partial_{\lambda} \text{ST} \left(\hat{\beta}^{(\lambda)} - \frac{1}{L} \nabla f(\hat{\beta}^{(\lambda)}), \frac{\lambda}{L} \right) \end{aligned}$$

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Key observation, if $\hat{\beta}_j^{(\lambda)} = 0$:

$$\partial_{\beta} \text{ST} \left(\hat{\beta}_j^{(\lambda)} - \frac{1}{L} \nabla_j f(\hat{\beta}^{(\lambda)}), \frac{\lambda}{L} \right) = 0 = \partial_{\lambda} \text{ST} \left(\hat{\beta}_j^{(\lambda)} - \frac{1}{L} \nabla_j f(\hat{\beta}^{(\lambda)}), \frac{\lambda}{L} \right)$$

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Key observation, if $\hat{\beta}_j^{(\lambda)} = 0$:

$$\partial_{\beta} \text{ST} \left(\hat{\beta}_j^{(\lambda)} - \frac{1}{L} \nabla_j f(\hat{\beta}^{(\lambda)}), \frac{\lambda}{L} \right) = 0 = \partial_{\lambda} \text{ST} \left(\hat{\beta}_j^{(\lambda)} - \frac{1}{L} \nabla_j f(\hat{\beta}^{(\lambda)}), \frac{\lambda}{L} \right)$$

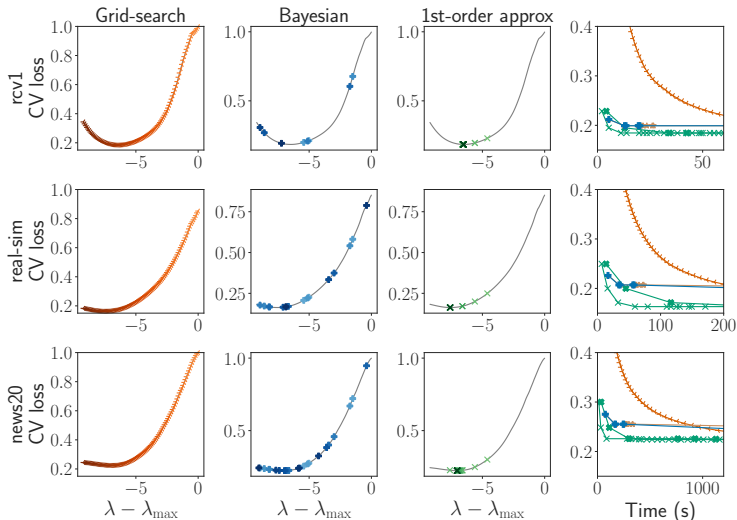
With $\mathcal{S} = \{j \in [p] : \hat{\beta}_j^{(\lambda)} = 0\}$ we have $\hat{\mathcal{J}}_{\mathcal{S}^c} = 0$

$$\hat{\mathcal{J}}_{\mathcal{S}} = \partial_{\beta} \text{ST}(\hat{\beta}^{(\lambda)} - \frac{1}{L} \nabla f(\hat{\beta}^{(\lambda)}), \frac{\lambda}{L})_{\mathcal{S}} \hat{\mathcal{J}}_{\mathcal{S}} + \partial_{\lambda} \text{ST}(\hat{\beta}_j^{(\lambda)} - \frac{1}{L} \nabla_j f(\hat{\beta}^{(\lambda)}), \frac{\lambda}{L})_{\mathcal{S}}$$

⁽¹⁾ Q. Bertrand, Q. Klopfenstein, M. Massias, et al. (2021). "Implicit differentiation for fast hyperparameter selection in non-smooth convex learning".
In: *Submitted to JMLR*

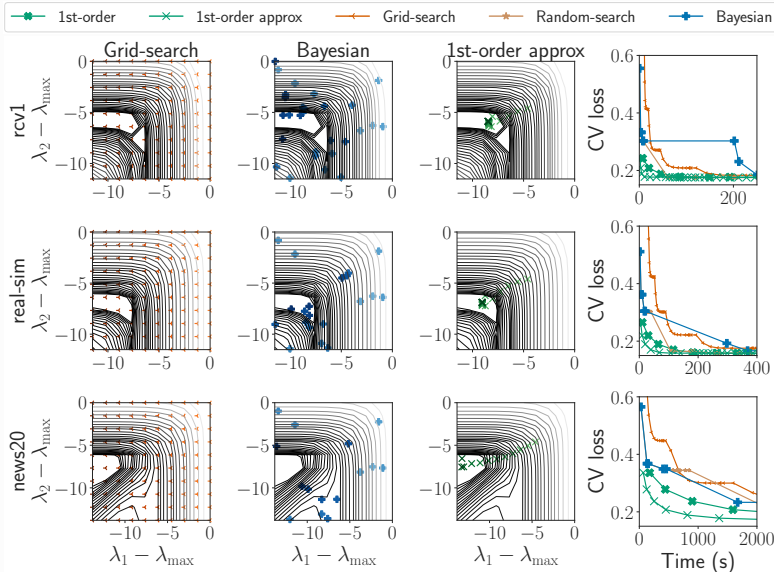
EXPERIMENTS I - LASSO CROSS-VALIDATION

● 1st-order
 x 1st-order approx
 + Grid-search
 * Random-search
 + Bayesian



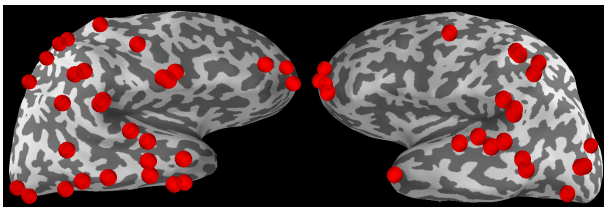
$$\arg \min_{\beta} \frac{1}{2n} \|y^{\text{train}} - X^{\text{train}} \beta\|^2 + e^{\lambda} \|\beta\|_1$$

EXPERIMENTS II - ENET CROSS-VALIDATION



$$\arg \min_{\beta} \frac{1}{2n} \|y^{\text{train}} - X^{\text{train}} \beta\|^2 + e^{\lambda_1} \|\beta\|_1 + \frac{e^{\lambda_2}}{2} \|\beta\|^2$$

- ▶ **Outer criterion:** FDMC SURE⁽¹⁾
- ▶ **Inner problems:** vanilla Lasso

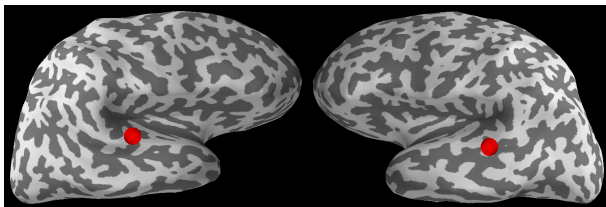


Real M/EEG data, vanilla Lasso (1 hyperparameter λ)

$$\arg \min_{\beta \in \mathbb{R}^p} \frac{1}{2n} \|y - X\beta\|_2^2 + e^\lambda \|\beta\|_1$$

⁽¹⁾ C.-A. Deledalle et al. (2014). "Stein Unbiased GrADient estimator of the Risk (SUGAR) for multiple parameter selection". In: *SIAM J. Imaging Sci.*

- ▶ **Outer criterion:** FDMC SURE⁽¹⁾
- ▶ **Inner problems:** weighted Lasso



Real M/EEG data, weighted Lasso (p hyperparameters)

$$\arg \min_{\beta \in \mathbb{R}^p} \frac{1}{2n} \|y - X\beta\|_2^2 + \sum_{j=1}^p e^{\lambda_j} |\beta_j|$$

⁽¹⁾ C.-A. Deledalle et al. (2014). "Stein Unbiased GrAdient estimator of the Risk (SUGAR) for multiple parameter selection". In: *SIAM J. Imaging Sci.*

- ▶ **Local linear convergence** of the Jacobian
- ▶ **Leverage sparsity** to speed up hypergradient computation
- ▶ Open source package

<https://github.com/QB3/sparse-ho>

[sparse-ho](#) [0.1.dev](#) [Examples](#) [API](#) [GitHub](#) [Site ▾](#)

sparse-ho

sparse-ho stands for "sparse hyperparameter optimization". This package implements efficient hyperparameter tuning for sparse machine learning models. It supports models such as the Lasso, the Weighted Lasso, multiclass sparse Logistic regression, SVM, etc.

Relying on a first order algorithm for bilevel optimization, *sparse-ho*'s performances scales gracefully with the number of hyperparameters to tune.

In order to benchmark performances, the package also implements alternatives such as forward or backward differentiation.

Documentation

Please visit '<https://qb3.github.io/sparse-ho>' for the latest version of the documentation.

Install

To run the code you first need to clone the repository, and then run, in the folder containing the `setup.py` file (root folder):

```
pip install -e .
```

*"An article about computational science in a scientific publication is **not** the scholarship itself, it is merely **advertising** of the scholarship. The actual scholarship is the complete software development environment and the complete set of instructions which generated the figures."*

J. B. Buckheit and D. L. Donoho⁽¹⁾

"All models are wrong, but some come with good open source implementation and good documentation: so use these."

A. Gramfort (circa 2015)

⁽¹⁾ J. B. Buckheit and D. L. Donoho (1995). "Wavelab and reproducible research". In: *Wavelets and statistics*. Springer, pp. 55–81.

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